ENVIRONMENTAL BIOTECHNOLOGY

(BT503)

QUIZ NO. 01

FINAL TERM

1. Capillary water maintain itself adsorbed in which of the following phase?
- **Solid**
- liquid
- Semi solid
- Gaseous

2. ----------techniques is the best suited to instances of relatively localized pollution within a site, typically in "hot spots" of medium to relatively high concentrations.
- Vitrification
- In-situ
- **Ex-situ**
- Solidification

3. A sludge with a solids load of 120KgSS/d and a solids concentration of 2.0% (20,000mg/L) will have a flow of?
- 4.0m 3/d
- 5.0m 3/d
- **6.0m 3/d**
- 7.0m 3/d

4. In a dewatering process the dry solid level in the dewatered sludge of filter press ----------.
- 30-45%
- 20-30%
- **25-40%**
- 80-90%

5. All of these are artificial processes of pathogen removal except?
- Chlorination
- **Land treatment**
- Ozonisation
- Membrane

6. For preliminary waste water treatment, all basic equipment are required except one?
- Grit chamber
- Screen
- **Aerator**
- Flow meter

7. Which of the following pathogen removal system does not need chemicals?
- **Membrane ponds**
- Chlorination
- Ozonisation

- UV radiation

8. Soil washing and thermal treatments are examples of ------- technology.

- **Intensive technologies**
- Extensive technologies
- Semi extensive
- None of these

9. Following chemicals are added in the sludge preparation process.

- Coagulants
- Sulphur
- Polyelectrolytes
- **Both a & c**

10. The process in which decomposition of aerobic organic matter is achieved through controlled conditions of temperature. water content and nutrients is known as-------

- **Compositing**
- Lime stabilization
- Percolation
- Filterable assembly

11. Facultative ponds require how much time for sludge removal from liquid phase?

- Weeks
- **Days**
- Months
- Decades

12. Which of the following method affects the reproduction of the pathogenic agents?

- Natural
- **UV radiation**
- Maturation ponds
- None

13. Which of the following sludge management stage accomplished through the removal of the biodegradable organic matter of the sludge?

- Thickening
- Dewatering
- **Stabilization**
- Conditioning

14. Disposal of sludge in the mining areas is known as --------.

- Sanitary landfill
- Maturation ponds
- **Land reclamation**
- Ocean disposal

15. ---------- is normally uses high temperature to fuse the contaminated materials.

- Aerators
- **Vitrification**
- Effluent recycle pump
- Incineration

16. Cropping of heavy metal accumulator plants occur in ---------- technology.

- Intensive technologies
- **Extensive technologies**
- Semi extensive
- None of the above

17. Which of the following is highly recommended system for pathogen removal?

- **Maturation ponds**
- Chlorination
- UV radiation
- Ozonisation

18. The design of sludge treatment and final disposal stages is based on which of the following?

- Sludge flow
- Dry solids load
- **Both**
- None

19. The main characteristic of --------- Methods are that the soil is removed from where it originally lay, for treatment?

- Chemical
- In-situ
- **Ex-situ**
- Physical

20. A sludge with a level of dry solids of 2% has water content of---------.

- 96%
- 97%
- **98%**
- 90%

21. A very effective method for the removal of disease causing organisms is?

- Maturation ponds
- Chlorination
- **Ozonisation**
- UV radiation

22. ---------- are well suited to large scale treatment where speed is not of the essence.

- Intensive technologies
- **Extensive technologies**
- Semi Extensive
- None

23. In-situ technique is suitable for ---------- contamination.

- **Widespread**
- Localized
- Organic
- Inorganic

24. Most frequently used unit for concentration of sludge is?

- Grams
- mg
- **mg/L**
- **%**

25. Among dewatering processes the process which is least used is called--.

- Strip method
- **Stabilization**
- Sludge lagoons
- Drying beds

26. Which of the following process does not introduce chemical products into liquid.

- **Membranes**
- UV radiation
- Maturation ponds
- Chlorination

27.